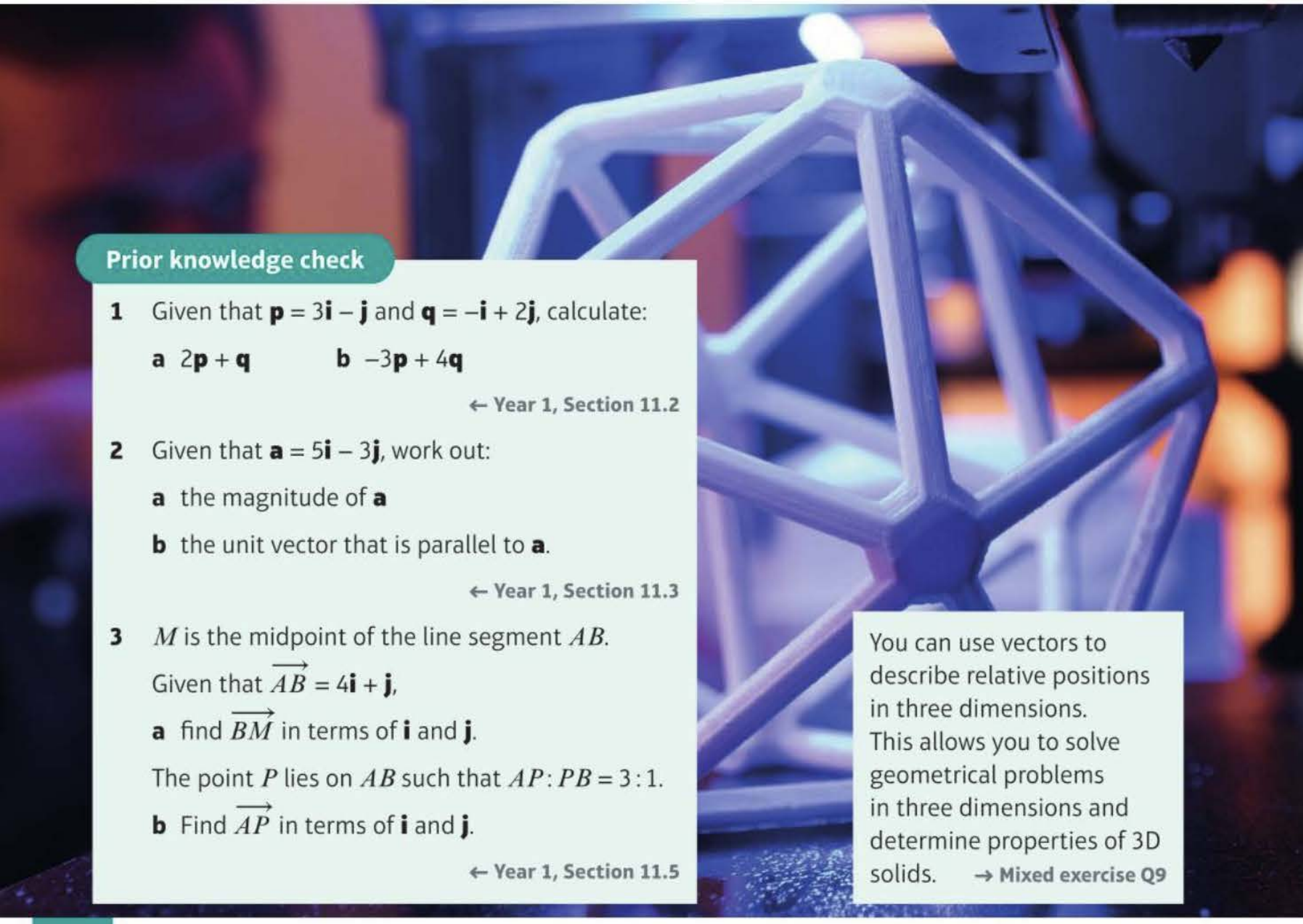
12 Vectors

Objectives

After completing this chapter you should be able to:

- Understand 3D Cartesian coordinates → pages 337–338
- Use vectors in three dimensions → pages 339–343
- Use vectors to solve geometric problems → pages 344–347
- Model 3D motion in mechanics with vectors → pages 347–349



Prior knowledge check

1 Given that $\mathbf{p} = 3\mathbf{i} - \mathbf{j}$ and $\mathbf{q} = -\mathbf{i} + 2\mathbf{j}$, calculate:

a $2\mathbf{p} + \mathbf{q}$ **b** $-3\mathbf{p} + 4\mathbf{q}$

← Year 1, Section 11.2

2 Given that $\mathbf{a} = 5\mathbf{i} - 3\mathbf{j}$, work out:

a the magnitude of $\mathbf{a}$

b the unit vector that is parallel to $\mathbf{a}$.

← Year 1, Section 11.3

3 M is the midpoint of the line segment AB.

Given that $\overrightarrow{AB} = 4\mathbf{i} + \mathbf{j}$,

a find $\overrightarrow{BM}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

The point P lies on AB such that $AP : PB = 3 : 1$.

b Find $\overrightarrow{AP}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

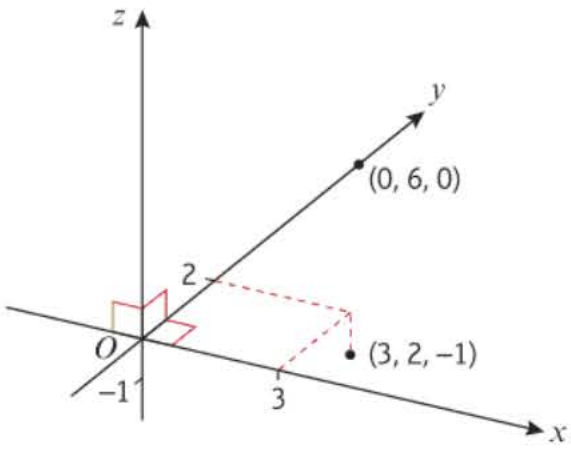
← Year 1, Section 11.5

You can use vectors to describe relative positions in three dimensions. This allows you to solve geometrical problems in three dimensions and determine properties of 3D solids. → Mixed exercise Q9

12.1 3D coordinates

Cartesian coordinate axes in three dimensions are usually called x-, y- and z-axes, each being at right angles to each of the others.

The coordinates of a point in three dimensions are written as (x, y, z).



Hint To visualise this, think of the x- and y-axes being drawn on a flat surface and the z-axis sticking up from the surface.

You can use Pythagoras' theorem in 3D to find distances on a 3D coordinate grid.

- **The distance from the origin to the point (x, y, z) is $\sqrt{x^2 + y^2 + z^2}$.**

Example 1

Find the distance from the origin to the point $P(4, -7, -1)$.

$$OP = \sqrt{4^2 + (-7)^2 + (-1)^2}$$
$$= \sqrt{16 + 49 + 1}$$
$$= \sqrt{66}$$

Substitute the values of x, y and z into the formula. You don't need to give units with distances on a coordinate grid.

You can also use Pythagoras' theorem to find the distance between two points.

- **The distance between the points (x_1, y_1, z_1) and (x_2, y_2, z_2) is**
$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

Example 2

Find the distance between the points $A(1, 3, 4)$ and $B(8, 6, -5)$, giving your answer to 1 d.p.

$$AB = \sqrt{(1 - 8)^2 + (3 - 6)^2 + (4 - (-5))^2}$$
$$= \sqrt{(-7)^2 + (-3)^2 + 9^2}$$
$$= \sqrt{49 + 9 + 81}$$
$$= \sqrt{139} = 11.8 \text{ (1 d.p.)}$$

Be careful with the signs – use brackets when you substitute.

Example 3

The coordinates of A and B are $(5, 0, 3)$ and $(4, 2, k)$ respectively.
Given that the distance from A to B is 3 units, find the possible values of k.

$$AB = \sqrt{(5-4)^2 + (0-2)^2 + (3-k)^2} = 3$$
$$\sqrt{1 + 4 + (9 - 6k + k^2)} = 3$$
$$1 + 4 + 9 - 6k + k^2 = 9$$
$$k^2 - 6k + 5 = 0$$
$$(k - 5)(k - 1) = 0$$
$$k = 1 \text{ or } k = 5$$

Problem-solving

Use Pythagoras' theorem to form a quadratic equation in k.

Square both sides of the equation.

Solve to find the two possible values of k.

Online Explore the solution to this example visually in 3D using GeoGebra.



Exercise 12A

1 Find the distance from the origin to the point $P(2, 8, -4)$.

2 Find the distance from the origin to the point $P(7, 7, 7)$.

3 Find the distance between A and B when they have the following coordinates:

a $A(3, 0, 5)$ and $B(1, -1, 8)$

b $A(8, 11, 8)$ and $B(-3, 1, 6)$

c $A(3, 5, -2)$ and $B(3, 10, 3)$

d $A(-1, -2, 5)$ and $B(4, -1, 3)$

(P) **4** The coordinates of A and B are $(7, -1, 2)$ and $(k, 0, 4)$ respectively.
Given that the distance from A to B is 3 units, find the possible values of k.

(P) **5** The coordinates of A and B are $(5, 3, -8)$ and $(1, k, -3)$ respectively.
Given that the distance from A to B is $3\sqrt{10}$ units, find the possible values of k.

Challenge

a The points $A(1, 3, -2)$, $B(1, 3, 4)$ and $C(7, -3, 4)$ are three vertices of a solid cube. Write down the coordinates of the remaining five vertices.

An ant walks from A to C along the surface of the cube.

b Determine the length of the shortest possible route the ant can take.

12.2 Vectors in 3D

You can use 3D vectors to describe position and displacement relative to the x-, y- and z-axes.
You can represent 3D vectors as column vectors or using the unit vectors **i**, **j** and **k**.

- **The unit vectors along the x-, y- and z-axes are denoted by i, j and k respectively.**

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

- **For any 3D vector $p\mathbf{i} + q\mathbf{j} + r\mathbf{k} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$**

Links 3D vectors obey all the same addition and scalar multiplication rules as 2D vectors.
← Year 1, Chapter 11

Example 4

Consider the points $A(1, 5, -2)$ and $B(0, -3, 7)$.

a Find the position vectors of A and B in **ijk** notation.

b Find the vector $\overrightarrow{AB}$ as a column vector.

a $\overrightarrow{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$, $\overrightarrow{OB} = -3\mathbf{j} + 7\mathbf{k}$

b $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

$$= \begin{pmatrix} 0 \\ -3 \\ 7 \end{pmatrix} - \begin{pmatrix} 1 \\ 5 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ -8 \\ 9 \end{pmatrix}$$

The position vector of a point is the vector from the origin to that point.

$\overrightarrow{OB}$ has no component in the **i** direction. You could write it as $0\mathbf{i} - 3\mathbf{j} + 7\mathbf{k}$.

When adding and subtracting vectors it is often easier to write them as column vectors.

Example 5

The vectors **a** and **b** are given as $\mathbf{a} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$.

Online Perform calculations on 3D vectors using your calculator.



a Find:

i $4\mathbf{a} + \mathbf{b}$ **ii** $2\mathbf{a} - 3\mathbf{b}$

b State with a reason whether each of these vectors is parallel to $4\mathbf{i} - 5\mathbf{k}$.

a i $4\mathbf{a} + \mathbf{b} = 4\begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$

$$= \begin{pmatrix} 8 \\ -12 \\ 20 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 12 \\ -14 \\ 20 \end{pmatrix}$$

Use the rules for scalar multiplication and addition of vectors:

$$\lambda\begin{pmatrix} p \\ q \\ r \end{pmatrix} = \begin{pmatrix} \lambda p \\ \lambda q \\ \lambda r \end{pmatrix} \text{ and } \begin{pmatrix} p \\ q \\ r \end{pmatrix} + \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} p+u \\ q+v \\ r+w \end{pmatrix}$$

ii $2\mathbf{a} - 3\mathbf{b} = 2\begin{pmatrix}2\\-3\\5\end{pmatrix} - 3\begin{pmatrix}4\\-2\\0\end{pmatrix}$

$= \begin{pmatrix}4\\-6\\10\end{pmatrix} - \begin{pmatrix}12\\-6\\0\end{pmatrix}$

$= \begin{pmatrix}-8\\0\\10\end{pmatrix}$

b i $4\mathbf{a} + \mathbf{b} = \begin{pmatrix}12\\-14\\20\end{pmatrix} = 3\begin{pmatrix}4\\\frac{-14}{3}\\\frac{20}{3}\end{pmatrix}$

which is not a multiple of $\begin{pmatrix}4\\0\\-5\end{pmatrix}$

$4\mathbf{a} + \mathbf{b}$ is not parallel to $4\mathbf{i} - 5\mathbf{k}$

Two vectors are parallel if one is a multiple of the other. Make the x-components the same and compare the y- and z-components with $4\mathbf{i} - 5\mathbf{k}$.

ii $2\mathbf{a} - 3\mathbf{b} = \begin{pmatrix}-8\\0\\10\end{pmatrix} = -2\begin{pmatrix}4\\0\\-5\end{pmatrix}$

which is a multiple of $\begin{pmatrix}4\\0\\-5\end{pmatrix}$

$2\mathbf{a} - 3\mathbf{b}$ is parallel to $4\mathbf{i} - 5\mathbf{k}$

Watch out $4\mathbf{i} - 5\mathbf{k} = 4\mathbf{i} + 0\mathbf{j} - 5\mathbf{k}$. Make sure you include a 0 in the **j**-component of the column vector.

Example 6

Find the magnitude of $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ and hence find $\hat{\mathbf{a}}$, the unit vector in the direction of $\mathbf{a}$.

The magnitude of $\mathbf{a}$ is given by

$|\mathbf{a}| = \sqrt{2^2 + (-1)^2 + 4^2}$

$= \sqrt{21}$

$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|} = \frac{1}{\sqrt{21}}(2\mathbf{i} - \mathbf{j} + 4\mathbf{k})$

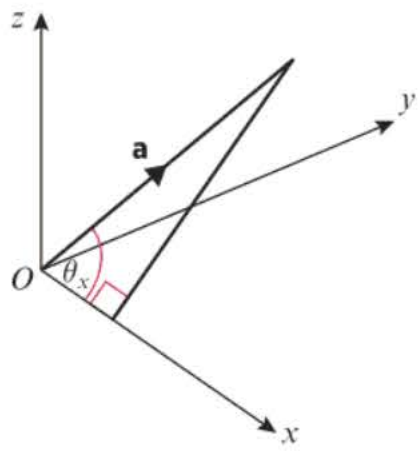
Use Pythagoras' theorem.

You could also write this as $\frac{2}{\sqrt{21}}\mathbf{i} - \frac{1}{\sqrt{21}}\mathbf{j} + \frac{4}{\sqrt{21}}\mathbf{k}$

Online Check your answer using the vector functions on your calculator.



You can find the angle between a given vector and any of the coordinate axes by considering the appropriate right-angled triangle.



- **If the vector $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ makes an angle θ_x with the positive x-axis then $\cos\theta_x = \frac{x}{|\mathbf{a}|}$ and similarly for the angles θ_y and θ_z**

Hint This rule also works with vectors in two dimensions.

Example 7

Find the angles that the vector $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$ makes with each of the positive coordinate axes to 1 d.p.

$|\mathbf{a}| = \sqrt{2^2 + (-3)^3 + (-1)^2} = \sqrt{4 + 9 + 1} = \sqrt{14}$

First find $|\mathbf{a}|$ since you will be using it three times.

$\cos\theta_x = \frac{x}{|\mathbf{a}|} = \frac{2}{\sqrt{14}} = 0.5345...$

$\theta_x = 57.7°$ (1 d.p.)

Write down at least 4 d.p., or use the answer button on your calculator to enter the exact value.

$\cos\theta_y = \frac{y}{|\mathbf{a}|} = \frac{-3}{\sqrt{14}} = -0.8017...$

$\theta_y = 143.3°$ (1 d.p.)

The formula also works with negative components. The y-component is negative, so the vector makes an obtuse angle with the positive y-axis.

$\cos\theta_z = \frac{z}{|\mathbf{a}|} = \frac{-1}{\sqrt{14}} = -0.2672...$

$\theta_z = 105.5°$ (1 d.p.)

Example 8

The points A and B have position vectors $4\mathbf{i} + 2\mathbf{j} + 7\mathbf{k}$ and $3\mathbf{i} + 4\mathbf{j} - \mathbf{k}$ relative to a fixed origin, O. Find $\overrightarrow{AB}$ and show that $\triangle OAB$ is isosceles.

$\overrightarrow{OA} = \mathbf{a} = \begin{pmatrix} 4 \\ 2 \\ 7 \end{pmatrix}$, $\overrightarrow{OB} = \mathbf{b} = \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix}$

Write down the position vectors of A and B.

$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 4 \\ 2 \\ 7 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ -8 \end{pmatrix}$

Use $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$.

$|\overrightarrow{AB}| = \sqrt{(-1)^2 + 2^2 + (-8)^2} = \sqrt{69}$

This is the length of the line segment AB.

$|\overrightarrow{OA}| = \sqrt{4^2 + 2^2 + 7^2} = \sqrt{69}$

$|\overrightarrow{OB}| = \sqrt{3^2 + 4^2 + (-1)^2} = \sqrt{26}$

Find the lengths of the other sides OA and OB of $\triangle OAB$.

So $\triangle OAB$ is isosceles, with $AB = OA$.

Online Explore the solution to this example visually in 3D using GeoGebra.

Exercise 12B

1 The vectors $\mathbf{a}$ and $\mathbf{b}$ are defined by $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ -3 \\ 5 \end{pmatrix}$.

a Find:

i $\mathbf{a} - \mathbf{b}$ **ii** $-\mathbf{a} + 3\mathbf{b}$

b State with a reason whether each of these vectors is parallel to $6\mathbf{i} - 10\mathbf{j} + 18\mathbf{k}$.

2 The vectors $\mathbf{a}$ and $\mathbf{b}$ are defined by $\mathbf{a} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ -2 \\ 4 \end{pmatrix}$.

Show that the vector $3\mathbf{a} + 2\mathbf{b}$ is parallel to $6\mathbf{i} + 4\mathbf{j} + 10\mathbf{k}$.

(P) **3** The vectors **a** and **b** are defined by $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$.

Given that $\mathbf{a} + 2\mathbf{b} = 5\mathbf{i} + 4\mathbf{j}$, find the values of p, q and r.

4 Find the magnitude of:

a $3\mathbf{i} + 5\mathbf{j} + \mathbf{k}$ **b** $4\mathbf{i} - 2\mathbf{k}$ **c** $\mathbf{i} + \mathbf{j} - \mathbf{k}$

d $5\mathbf{i} - 9\mathbf{j} - 8\mathbf{k}$ **e** $\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}$

5 Given that $\mathbf{p} = \begin{pmatrix} 5 \\ 0 \\ 2 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} 7 \\ -4 \\ 2 \end{pmatrix}$, find in column vector form:

a $\mathbf{p} + \mathbf{q}$ **b** $\mathbf{q} - \mathbf{r}$ **c** $\mathbf{p} + \mathbf{q} + \mathbf{r}$

d $3\mathbf{p} - \mathbf{r}$ **e** $\mathbf{p} - 2\mathbf{q} + \mathbf{r}$

6 The position vector of the point A is $2\mathbf{i} - 7\mathbf{j} + 3\mathbf{k}$ and $\overrightarrow{AB} = 5\mathbf{i} + 4\mathbf{j} - \mathbf{k}$.
Find the position vector of the point B.

(P) **7** Given that $\mathbf{a} = t\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, and that $|\mathbf{a}| = 7$, find the possible values of t.

(P) **8** Given that $\mathbf{a} = 5t\mathbf{i} + 2t\mathbf{j} + t\mathbf{k}$, and that $|\mathbf{a}| = 3\sqrt{10}$, find the possible values of t.

9 The points A, B and C have coordinates $(2, 1, 4)$, $(3, -2, 4)$ and $(-1, 2, 2)$.

a Find, in terms of **i**, **j** and **k**:

i the position vectors of A, B and C

ii $\overrightarrow{AC}$

b Find the exact value of:

i $|\overrightarrow{AC}|$

ii $|\overrightarrow{OC}|$

10 P is the point $(3, 0, 7)$ and Q is the point $(-1, 3, -5)$. Find:

a the vector $\overrightarrow{PQ}$

b the distance between P and Q

c the unit vector in the direction of $\overrightarrow{PQ}$.

11 $\overrightarrow{OA}$ is the vector $4\mathbf{i} - \mathbf{j} - 2\mathbf{k}$ and $\overrightarrow{OB}$ is the vector $-2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$. Find:

a the vector $\overrightarrow{AB}$

b the distance between A and B

c the unit vector in the direction of $\overrightarrow{AB}$.

12 Find the unit vector in the direction of each of the following vectors.

a $\mathbf{p} = \begin{pmatrix} 3 \\ -4 \\ -2 \end{pmatrix}$ **b** $\mathbf{q} = \begin{pmatrix} \sqrt{2} \\ -4 \\ -\sqrt{7} \end{pmatrix}$ **c** $\mathbf{r} = \begin{pmatrix} \sqrt{5} \\ -2\sqrt{2} \\ -\sqrt{3} \end{pmatrix}$

E/P **13** The points A, B and C have position vectors $\begin{pmatrix} 8 \\ -7 \\ 4 \end{pmatrix}$, $\begin{pmatrix} 8 \\ -3 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 12 \\ -6 \\ 3 \end{pmatrix}$ respectively.

a Find the vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{BC}$. **(3 marks)**

b Find $|\overrightarrow{AB}|$, $|\overrightarrow{AC}|$ and $|\overrightarrow{BC}|$ giving your answers in exact form. **(6 marks)**

c Describe triangle ABC. **(1 mark)**

E **14** A is the point $(3, 4, 8)$, B is the point $(1, -2, 5)$ and C is the point $(7, -5, 7)$.

a Find the vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{BC}$. **(3 marks)**

b Hence find the lengths of the sides of triangle ABC. **(6 marks)**

c Given that angle $ABC = 90°$ find the size of angle BAC. **(2 marks)**

15 For each of the given vectors,

a $-\mathbf{i} + 7\mathbf{j} + \mathbf{k}$ **b** $\begin{pmatrix} 3 \\ 4 \\ 7 \end{pmatrix}$ **c** $\begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix}$

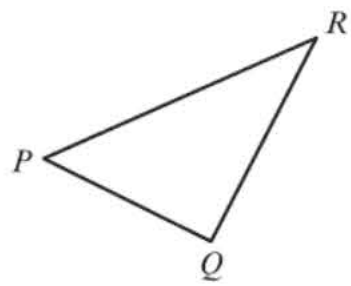
find the angle made by the vector with:

i the positive x-axis **ii** the positive y-axis **iii** the positive z-axis

P **16** A scalene triangle has the coordinates $(2, 0, 0)$, $(5, 0, 0)$ and $(4, 2, 3)$. Work out the area of the triangle.

E/P **17** The diagram shows the triangle PQR.

Given that $\overrightarrow{PQ} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\overrightarrow{QR} = -2\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$, show that $\angle PQR = 78.5°$ to 1 d.p.



(5 marks)

Challenge

Find the acute angle that the vector $\mathbf{a} = -2\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}$ makes with the x-y plane. Give your answer to 1 d.p.

12.3 Solving geometric problems

You need to be able to solve geometric problems involving vectors in three dimensions.

Example 9

A, B, C and D are the points $(2, -5, -8)$, $(1, -7, -3)$, $(0, 15, -10)$ and $(2, 19, -20)$ respectively.

a Find $\overrightarrow{AB}$ and $\overrightarrow{DC}$, giving your answers in the form $p\mathbf{i} + q\mathbf{j} + r\mathbf{k}$.

b Show that the lines AB and DC are parallel and that $\overrightarrow{DC} = 2\overrightarrow{AB}$.

c Hence describe the quadrilateral $ABCD$.

a $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

$= (\mathbf{i} - 7\mathbf{j} - 3\mathbf{k}) - (2\mathbf{i} - 5\mathbf{j} - 8\mathbf{k})$

$= -\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$

$\overrightarrow{DC} = \overrightarrow{OC} - \overrightarrow{OD}$

$= (15\mathbf{j} - 10\mathbf{k}) - (2\mathbf{i} + 19\mathbf{j} - 20\mathbf{k})$

$= -2\mathbf{i} - 4\mathbf{j} + 10\mathbf{k}$

b $2\overrightarrow{AB} = 2(-\mathbf{i} - 2\mathbf{j} + 5\mathbf{k})$

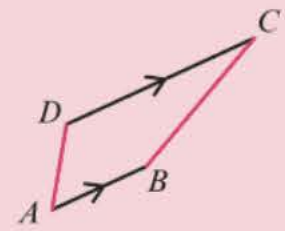
$= -2\mathbf{i} - 4\mathbf{j} + 10\mathbf{k} = \overrightarrow{DC}$

So AB is parallel to DC and half as long.

c There are two unequal parallel sides, so $ABCD$ is a trapezium.

Watch out AB refers to the **line segment** between A and B (or its length), whereas $\overrightarrow{AB}$ refers to the **vector** from A to B. Note that $AB = BA$ but $\overrightarrow{AB} \neq \overrightarrow{BA}$.

Problem-solving If you can't work out what shape it is, draw a sketch showing AB and DC.

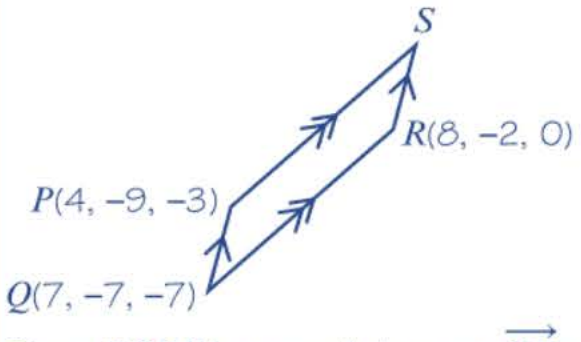


Online Explore the solution to this example visually in 3D using GeoGebra.



Example 10

P, Q and R are the points $(4, -9, -3)$, $(7, -7, -7)$ and $(8, -2, -0)$ respectively. Find the coordinates of the point S so that $PQRS$ forms a parallelogram.



Since $PQRS$ is a parallelogram, $\overrightarrow{QP} = \overrightarrow{RS}$.

So $\overrightarrow{OS} = \overrightarrow{OR} + \overrightarrow{RS} = \overrightarrow{OR} + \overrightarrow{QP}$

$\overrightarrow{OR} = \begin{pmatrix} 8 \\ -2 \\ 0 \end{pmatrix}$ and

$\overrightarrow{QP} = \overrightarrow{OP} - \overrightarrow{OQ} = \begin{pmatrix} 4 \\ -9 \\ -3 \end{pmatrix} - \begin{pmatrix} 7 \\ -7 \\ -7 \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \\ 4 \end{pmatrix}$

So $\overrightarrow{OS} = \begin{pmatrix} 8 \\ -2 \\ 0 \end{pmatrix} + \begin{pmatrix} -3 \\ -2 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 4 \end{pmatrix}$

which means that S is the point $(5, -4, 4)$

Draw a sketch. The vertices in a 2D shape are given in order, either clockwise or anticlockwise. It doesn't matter if the positions on the sketch don't correspond to the real positions in 3D – it is still a helpful way to visualise the problem.

You could also go from O to S via P:

$\overrightarrow{OS} = \overrightarrow{OP} + \overrightarrow{PS}$

$= \overrightarrow{OP} + \overrightarrow{QR}$

In two dimensions you saw that if **a** and **b** are two non-parallel vectors and $p\mathbf{a} + q\mathbf{b} = r\mathbf{a} + s\mathbf{b}$ then $p = r$ and $q = s$. In other words, in two dimensions with two vectors you can **compare coefficients** on both sides of an equation. In three dimensions you have to extend this rule:

- **If a, b and c are vectors in three dimensions which do not all lie on the same plane then you can compare their coefficients on both sides of an equation.**

Notation **Coplanar vectors** are vectors which are in the same plane.
Non-coplanar vectors are vectors which are **not** in the same plane.

In particular, since the vectors **i**, **j** and **k** are non-coplanar, if $p\mathbf{i} + q\mathbf{j} + r\mathbf{k} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$ then $p = u$, $q = v$ and $r = w$.

Example 11

Given that $3\mathbf{i} + (p + 2)\mathbf{j} + 120\mathbf{k} = p\mathbf{i} - q\mathbf{j} + 4pqr\mathbf{k}$, find the values of p, q and r.

Comparing coefficients of **i** gives $p = 3$.

Comparing coefficients of **j** gives $p + 2 = -q$ so $q = -(3 + 2) = -5$.

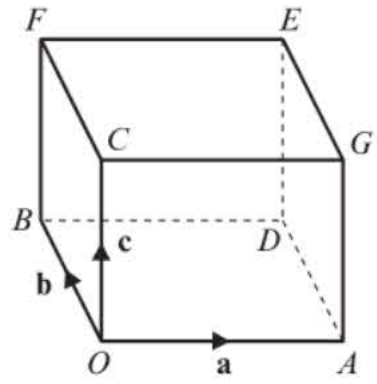
Comparing coefficients of **k** gives $120 = 4pqr$ so $r = \dfrac{120}{4 \times 3 \times (-5)} = -2$

Since **i**, **j** and **k** do not lie in the same plane you can compare coefficients.

When comparing coefficients like this just write the coefficients. For example, write $3 = p$, not $3\mathbf{i} = p\mathbf{i}$.

Example 12

The diagram shows a cuboid whose vertices are O, A, B, C, D, E, F and G. Vectors **a**, **b** and **c** are the position vectors of the vertices A, B and C respectively. Prove that the diagonals OE and BG bisect each other.



Hint Bisect means 'cut into two equal parts'. In this case you need to prove that **both** diagonals are bisected.

Problem-solving

If there is a point of intersection, H, it must lie on both diagonals. You can reach H directly from O (travelling along OE), or by first travelling to B then travelling along BG. Use this to write two expressions for $\overrightarrow{OH}$.

Suppose there is a point of intersection, H, of OE and BG.

$\overrightarrow{OH} = r\overrightarrow{OE}$ for some scalar r.

But $\overrightarrow{OH} = \overrightarrow{OB} + \overrightarrow{BH}$ and $\overrightarrow{BH} = s\overrightarrow{BG}$ for some scalar s, so $\overrightarrow{OH} = \overrightarrow{OB} + s\overrightarrow{BG}$.

H lies on the line OE, so $\overrightarrow{OH}$ must be some scalar multiple of $\overrightarrow{OE}$.

Use the fact that H lies on both diagonals to find two different expression for $\overrightarrow{OH}$. You can equate these expressions and compare coefficients.

So $r\overrightarrow{OE} = \overrightarrow{OB} + s\overrightarrow{BG}$ (1)

Now $\overrightarrow{OE} = \overrightarrow{OA} + \overrightarrow{AD} + \overrightarrow{DE} = \mathbf{a} + \mathbf{b} + \mathbf{c}$,

$\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{BG} = \overrightarrow{OG} - \overrightarrow{OB} = \mathbf{a} + \mathbf{c} - \mathbf{b}$

So (1) becomes $r(\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{b} + s(\mathbf{a} + \mathbf{c} - \mathbf{b})$

Comparing coefficients in **a** and **b** gives $r = s$ and $r = 1 - s$

Solving simultaneously gives $r = s = \frac{1}{2}$

These solutions also satisfy the coefficients of **c** so the lines do intersect at H.

$OH = \frac{1}{2}OE$ so H bisects OE.

$BH = \frac{1}{2}BG$ so H bisects BG, as required.

a, **b** and **c** are three non-coplanar vectors so you can compare coefficients.

In order for the lines to intersect, the values of r and s must satisfy equation (1) completely:

$\frac{1}{2}(\mathbf{a} + \mathbf{b} + \mathbf{c}) = \mathbf{b} + \frac{1}{2}(\mathbf{a} + \mathbf{c} - \mathbf{b})$

The coefficients of **a**, **b** and **c** all match so both ways of writing the vector OH are identical.

Vector proofs such as this one often avoid any coordinate geometry, which tends to be messy and complicated, especially in three dimensions.

Exercise 12C

(P) **1** The points A, B and C have position vectors $\begin{pmatrix}1\\-4\\8\end{pmatrix}$, $\begin{pmatrix}4\\4\\7\end{pmatrix}$ and $\begin{pmatrix}10\\0\\30\end{pmatrix}$ relative to a fixed origin, O.

a Show that:

i $|\overrightarrow{OA}| = |\overrightarrow{OB}|$ **ii** $|\overrightarrow{AC}| = |\overrightarrow{BC}|$

b Hence describe the quadrilateral $OACB$.

(P) **2** The points A, B and C have coordinates (2, 1, 5), (4, 4, 3) and (2, 7, 5) respectively.

a Show that triangle ABC is isosceles.

b Find the area of triangle ABC.

c Find a point D such that $ABCD$ is a parallelogram.

(P) **3** The points A, B, C and D have coordinates (7, 12, −1), (11, 2, −9), (14, −14, 3) and (8, 1, 15) respectively.

a Show that AB and CD are parallel, and find the ratio $AB : CD$ in its simplest form.

b Hence describe the quadrilateral $ABCD$.

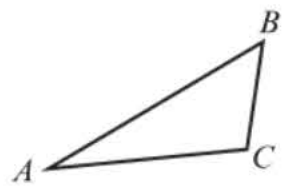
(P) **4** Given that $(3a + b)\mathbf{i} + \mathbf{j} + ac\mathbf{k} = 7\mathbf{i} - b\mathbf{j} + 4\mathbf{k}$, find the values of a, b and c.

(P) **5** The points A and B have position vectors $10\mathbf{i} - 23\mathbf{j} + 10\mathbf{k}$ and $p\mathbf{i} + 14\mathbf{j} - 22\mathbf{k}$ respectively, relative to a fixed origin O, where p is a constant.

Given that $\triangle OAB$ is isosceles, find **three** possible positions of point B.

(E/P) **6** The diagram shows a triangle ABC.

Given that $\overrightarrow{AB} = 7\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\overrightarrow{BC} = -\mathbf{i} + 5\mathbf{k}$

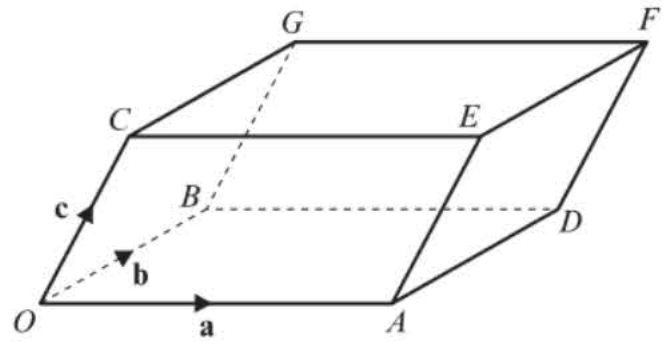


a find the area of triangle ABC. **(7 marks)**

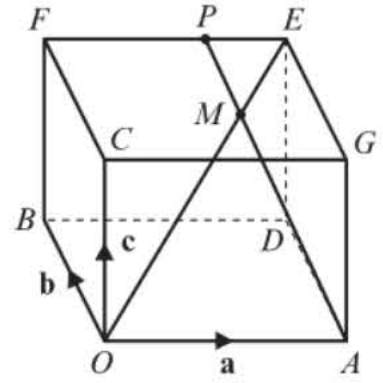
The point D is such that $\overrightarrow{AD} = 3\overrightarrow{AB}$, and the point E is such that $\overrightarrow{AE} = 3\overrightarrow{AC}$.

b Find the area of triangle ADE. **(2 marks)**

(P) **7** A parallelepiped is a three-dimensional figure formed by six parallelograms. The diagram shows a parallelepiped with vertices O, A, B, C, D, E, F, and G. **a**, **b** and **c** are the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ respectively. Prove that the diagonals OF and AG bisect each other.



(P) **8** The diagram shows a cuboid whose vertices are O, A, B, C, D, E, F and G. **a**, **b** and **c** are the position vectors of the vertices A, B and C respectively. The point M lies on OE such that $OM : ME = 3 : 1$. The straight line AP passes through point M. Given that $AM : MP = 3 : 1$, prove that P lies on the line EF and find the ratio $FP : PE$.



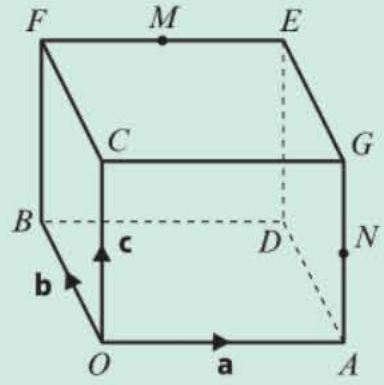
Challenge

1 **a**, **b** and **c** are the vectors $\begin{pmatrix}1\\0\\4\end{pmatrix}$, $\begin{pmatrix}2\\0\\-3\end{pmatrix}$ and $\begin{pmatrix}-5\\3\\1\end{pmatrix}$ respectively. Find scalars p, q and r such that $p\mathbf{a} + q\mathbf{b} + r\mathbf{c} = \begin{pmatrix}28\\-12\\-4\end{pmatrix}$

2 The diagram shows a cuboid with vertices O, A, B, C, D, E, F and G. M is the midpoint of FE and N is the midpoint of AG.

a, **b** and **c** are the position vectors of the vertices A, B and C respectively.

Prove that the lines OM and BN trisect the diagonal AF.



Hint Trisect means divide into three equal parts.

12.4 Application to mechanics

3D vectors can be used to model problems in mechanics in the same way as you have previously used 2D vectors.

Example 13

A particle of mass 0.5 kg is acted on by three forces:

$\mathbf{F}_1 = (2\mathbf{i} - \mathbf{j} + 2\mathbf{k})$ N

$\mathbf{F}_2 = (-\mathbf{i} + 3\mathbf{j} - 3\mathbf{k})$ N

$\mathbf{F}_3 = (4\mathbf{i} - 3\mathbf{j} - 2\mathbf{k})$ N

a Find the resultant force $\mathbf{R}$ acting on the particle.

b Find the acceleration of the particle, giving your answer in the form $(p\mathbf{i} + q\mathbf{j} + r\mathbf{k})$ m s^{-2}.

c Find the magnitude of the acceleration.

Given that the particle starts at rest,

d find the distance travelled by the particle in the first 6 seconds of its motion.

a $\begin{pmatrix}2\\-1\\2\end{pmatrix} + \begin{pmatrix}-1\\3\\-3\end{pmatrix} + \begin{pmatrix}4\\-3\\-2\end{pmatrix} = \begin{pmatrix}5\\-1\\-3\end{pmatrix}$

$\mathbf{R} = (5\mathbf{i} - \mathbf{j} - 3\mathbf{k})\,\text{N}$

b $\mathbf{F} = m\mathbf{a}$

$(5\mathbf{i} - \mathbf{j} - 3\mathbf{k}) = 0.5\,\mathbf{a}$

$\mathbf{a} = (10\mathbf{i} - 2\mathbf{j} - 6\mathbf{k})\,\text{m s}^{-2}$

c $|\mathbf{a}| = \sqrt{10^2 + (-2)^2 + (-6)^2} = \sqrt{140}\ \text{m s}^{-2}$

d $u = 0,\ a = \sqrt{140}\ \text{m s}^{-2},\ t = 6\text{s},\ s = ?$

$s = ut + \frac{1}{2}at^2$

$= \frac{1}{2} \times \sqrt{140} \times 6^2$

$= 36\sqrt{35}\ \text{m}$

$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$

It is easier to add vectors in column vector form.

Use the vector form of $\mathbf{F} = m\mathbf{a}$. This is Newton's second law of motion.

← Statistics and Mechanics Year 1, Chapter 10

Because the particle starts from rest the acceleration acts in the same direction as the motion of the particle. So it is moving in a straight line with constant acceleration. You can use the *suvat* formulae to find the distance travelled.

← Statistics and Mechanics Year 1, Chapter 9

Exercise 12D

1 A particle is acted upon by forces of $(3\mathbf{i} - 2\mathbf{j} + \mathbf{k})\,\text{N}$, $(7\mathbf{i} + 4\mathbf{j} + 3\mathbf{k})\,\text{N}$ and $(-5\mathbf{i} - 3\mathbf{j})\,\text{N}$.

a Work out the resultant force $\mathbf{R}$.

b Find the exact magnitude of the resultant force.

(P) **2** A particle, initially at rest, is acted upon by a force that causes the particle to accelerate at $(4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})\,\text{m s}^{-2}$ for 2 seconds. Work out the distance travelled by the particle.

3 A body of mass 4 kg is moving with a constant velocity when it is acted upon by a force of $(2\mathbf{i} - 5\mathbf{j} + 3\mathbf{k})\,\text{N}$.

a Find the acceleration of the body while the force acts.

b Find the magnitude of this acceleration to 3 s.f.

(P) **4** A particle of mass 6 kg is acted on by two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$. Given that $\mathbf{F}_1 = (7\mathbf{i} + 3\mathbf{j} + \mathbf{k})\,\text{N}$, and that the particle is accelerating at $(2\mathbf{i} - \mathbf{k})\,\text{m s}^{-2}$, find $\mathbf{F}_2$, giving your answer in the form $(p\mathbf{i} + q\mathbf{j} + r\mathbf{k})\,\text{N}$.

(P) **5** A particle of mass 2 kg is in static equilibrium and is acted upon by three forces:

$\mathbf{F}_1 = (\mathbf{i} - \mathbf{j} - 2\mathbf{k})\,\text{N}$

$\mathbf{F}_2 = (-\mathbf{i} + 3\mathbf{j} + b\mathbf{k})\,\text{N}$

$\mathbf{F}_3 = (a\mathbf{j} - 2\mathbf{k})\,\text{N}$

a Find the values of the constants a and b.

$\mathbf{F}_2$ is removed. Work out:

b the value of the resultant force $\mathbf{R}$

c the acceleration of the particle, giving your answer in the form $(p\mathbf{i} + q\mathbf{j} + r\mathbf{k})\,\text{m s}^{-2}$

d the magnitude of this acceleration

e the angle the acceleration vector makes with the unit vector $\mathbf{j}$.

6 In this question **i** and **j** are the unit vectors due east and north, and **k** is the unit vector vertically upwards. An aeroplane of mass 1200 kg is initially in level flight. The forces of thrust **T**, lift **L**, and the combined forces of wind and air resistance **F**, acting on the aeroplane are modelled as:

$\mathbf{T} = 2800\mathbf{i} - 1800\mathbf{j} + 300\mathbf{k}$
$\mathbf{L} = 11\,000\mathbf{k}$
$\mathbf{F} = -900\mathbf{i} + 500\mathbf{j}$

a Taking $g = 9.8\ \text{m s}^{-2}$, find the magnitude of the acceleration of the aeroplane.

b Determine whether the aeroplane is ascending or descending, and find the size of the obtuse angle its acceleration makes with the vector **k**.

Mixed exercise 12

(P) **1** The points $A(2, 7, 3)$ and $B(4, 3, 5)$ are joined to form the line segment AB. The point M is the midpoint of AB. Find the distance from M to the point $C(5, 8, 7)$.

(P) **2** The coordinates of P and Q are $(2, 3, a)$ and $(a - 2, 6, 7)$. Given that the distance from P to Q is $\sqrt{14}$, find the possible values of a.

(P) **3** $\overrightarrow{AB}$ is the vector $-3\mathbf{i} + t\mathbf{j} + 5\mathbf{k}$, where $t > 0$. Given that $|\overrightarrow{AB}| = 5\sqrt{2}$, show that $\overrightarrow{AB}$ is parallel to $6\mathbf{i} - 8\mathbf{j} - \frac{5}{2}t\mathbf{k}$.

(P) **4** P is the point $(5, 6, -2)$, Q is the point $(2, -2, 1)$ and R is the point $(2, -3, 6)$.

a Find the vectors $\overrightarrow{PQ}$, $\overrightarrow{PR}$ and $\overrightarrow{QR}$.

b Hence, or otherwise, find the area of triangle PQR.

(E/P) **5** The points D, E and F have position vectors $\begin{pmatrix}1\\0\\0\end{pmatrix}$, $\begin{pmatrix}5\\3\\4\end{pmatrix}$ and $\begin{pmatrix}2\\-1\\8\end{pmatrix}$ respectively.

a Find the vectors $\overrightarrow{DE}$, $\overrightarrow{EF}$ and $\overrightarrow{FD}$. **(3 marks)**

b Find $|\overrightarrow{DE}|$, $|\overrightarrow{EF}|$ and $|\overrightarrow{FD}|$ giving your answers in exact form. **(6 marks)**

c Describe triangle DEF. **(1 mark)**

(E) **6** P is the point $(-6, 2, 1)$, Q is the point $(3, -2, 1)$ and R is the point $(1, 3, -2)$.

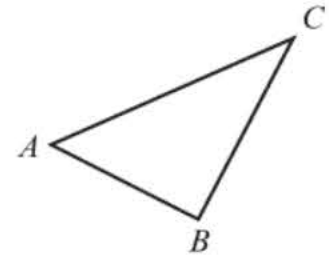
a Find the vectors $\overrightarrow{PQ}$, $\overrightarrow{PR}$ and $\overrightarrow{QR}$. **(3 marks)**

b Hence find the lengths of the sides of triangle PQR. **(6 marks)**

c Given that angle $QRP = 90°$ find the size of angle PQR. **(2 marks)**

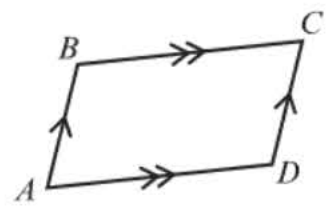
(E/P) **7** The diagram shows the triangle ABC.

Given that $\overrightarrow{AB} = -\mathbf{i} + \mathbf{j}$ and $\overrightarrow{BC} = \mathbf{i} - 3\mathbf{j} + \mathbf{k}$,
find $\angle ABC$ to 1 d.p.



(5 marks)

E/P **8** The diagram shows the quadrilateral $ABCD$.



Given that $\overrightarrow{AB} = \begin{pmatrix} 6 \\ -2 \\ 11 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} 15 \\ 8 \\ 5 \end{pmatrix}$, find the area of the quadrilateral. **(7 marks)**

P **9** A is the point $(2, 3, -2)$, B is the point $(0, -2, 1)$ and C is the point $(4, -2, -5)$. When A is reflected in the line BC it is mapped to the point D.

a Work out the coordinates of the point D.

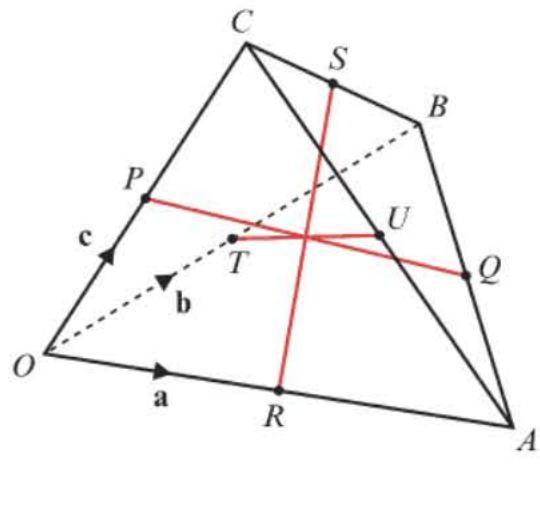
b Give the mathematical name for the shape $ABCD$.

c Work out the area of $ABCD$.

P **10** The diagram shows a tetrahedron $OABC$. **a**, **b** and **c** are the position vectors of A, B and C respectively.

P, Q, R, S, T and U are the midpoints of OC, AB, OA, BC, OB and AC respectively.

Prove that the line segments PQ, RS and TU meet at a point and bisect each other.

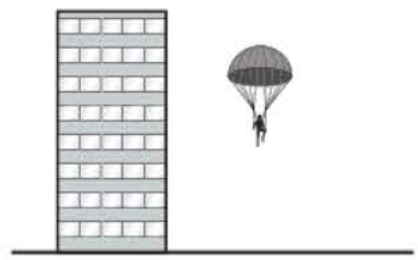


P **11** A particle of mass 2 kg is acted upon by three forces:

$\mathbf{F}_1 = (b\mathbf{i} + 2\mathbf{j} + \mathbf{k})$ N
$\mathbf{F}_2 = (3\mathbf{i} - b\mathbf{j} + 2\mathbf{k})$ N
$\mathbf{F}_3 = (-2\mathbf{i} + 2\mathbf{j} + (4 - b)\mathbf{k})$ N

Given that the particle accelerates at $3.5\,\text{m s}^{-2}$, work out the possible values of b.

P **12** In this question **i** and **j** are the unit vectors due east and due north respectively, and **k** is the unit vector acting vertically upwards.



A BASE jumper descending with a parachute is modelled as a particle of mass 50 kg subject to forces describing the wind, **W**, and air resistance, **F**, where:

$\mathbf{W} = (20\mathbf{i} + 16\mathbf{j})$ N
$\mathbf{F} = (-4\mathbf{i} - 3\mathbf{j} + 450\mathbf{k})$ N

a With reference to the model, suggest a reason why the **k** component of **F** is greater than the other components.

b Taking $g = 9.8\,\text{m s}^{-2}$, find the resultant force acting on the BASE jumper.

c Given that the BASE jumper starts from rest and travels a distance of 180 m before landing, find the total time of the descent.

Challenge

A student writes the following hypothesis:

> If **a**, **b** and **c** are three non-parallel vectors in three dimensions, then
> $p\mathbf{a} + q\mathbf{b} + r\mathbf{c} = s\mathbf{a} + t\mathbf{b} + u\mathbf{c} \Rightarrow p = s,\ q = t$ and $r = u$

Show, by means of a counter-example, that this hypothesis is not true.

Summary of key points

1 The distance from the origin to the point (x, y, z) is $\sqrt{x^2 + y^2 + z^2}$

2 The distance between the points (x_1, y_1, z_1) and (x_2, y_2, z_2) is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

3 The unit vectors along the x-, y- and z-axes are denoted by **i**, **j** and **k** respectively.

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Any 3D vector can be written in column form as $p\mathbf{i} + q\mathbf{j} + r\mathbf{k} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$

4 If the vector $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ makes an angle θ_x with the positive x-axis then $\cos\theta_x = \dfrac{x}{|\mathbf{a}|}$ and similarly for the angles θ_y and θ_z.

5 If **a**, **b** and **c** are vectors in three dimensions which do not all lie in the same plane then you can compare their coefficients on both sides of an equation.